

Conjecture on tightness of the upper bound for all $r \geq 3$

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

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Summary and provenance

Catalog id	2105.15195__00
Catalog entry	Conjecture on tightness of the upper bound for all $r \geq 3$
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	The upper logarithmic density of monochromatic subset sums
Model verdict	proved (confidence: high)
Model's one-line claim	A prime-based reduction closes the interlacing gap and proves $c_r = b_0^{(r-1)/(2r)}$ for every $r \geq 2$.
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2105.15195__00/output.md
Independent review	None
Model's caveats	The intended denominator is $2r b_0^{-r}$; the literal $2^r b_0^{-r}$ transcription is false for $r \geq 3$.

Problem statement

No extracted statement was stored in the artifact metadata.

Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

Exact value of the upper logarithmic density constant

For $A \subseteq \mathbb{N}$, write

$$\Sigma(A) = \left\{ \sum_{a \in F} a : \emptyset \neq F \subseteq A, F \text{ finite} \right\},$$

and put

$$\bar{d}_{\log}(S) = \limsup_{x \rightarrow \infty} \frac{1}{\log x} \sum_{\substack{n \leq x \\ n \in S}} \frac{1}{n}.$$

Thus the constant in question is

$$c_r = \inf_{\mathbb{N} = A_1 \cup \dots \cup A_r} \max_{i \in [r]} \bar{d}_{\log}(\Sigma(A_i)).$$

The key addition to the previous attempt is an additive lemma for **dense sets of primes**. It provides the required intervals of subset sums without requiring any monochromatic interval of integers. I also give a complete proof of the geometric lemma, including its boundary accounting.

Theorem

Let $r \geq 2$, and let $b > 1$ be the unique root of

$$b^r - 2rb + r - 1 = 0.$$

Then

$$c_r = \frac{b^{r-1}}{2r} = \frac{1 - \frac{1}{2b}}{1 - b^{-r}} = \left(1 - \frac{1}{2b}\right) \left(1 + \frac{1}{2rb - r}\right). \quad (1)$$

In particular, this proves the intended tightness conjecture. The distinction between $2r$ and 2^r is substantive; it is addressed explicitly at the end.

1. A geometric covering lemma

For $r \geq 2$ and $\lambda > 1$, define

$$D_{r,\lambda}(q) = \frac{1 - \frac{1}{\lambda q}}{1 - q^{-r}}, \quad q > 1,$$

and

$$C_{r,\lambda} = \inf_{q > 1} D_{r,\lambda}(q). \quad (2)$$

Notice that $0 < C_{r,\lambda} < 1$. Positivity follows from $D_{r,\lambda}(q) \geq 1 - 1/\lambda$, while $D_{r,\lambda}(q) < 1$ whenever $q^{r-1} > \lambda$.

Lemma 1 Suppose

$$0 < t_0 < t_1 < \dots, \quad t_k \longrightarrow \infty,$$

and assign each $k \geq 0$ a label $\gamma_k \in [r]$. Set

$$J_k = [t_k, \lambda t_{k+1}], \quad U_i = \bigcup_{\gamma_k = i} J_k.$$

Then

$$\max_{i \in [r]} \limsup_{T \rightarrow \infty} \frac{|U_i \cap [0, T]|}{T} \geq C_{r,\lambda}, \quad (3)$$

where $|\cdot|$ is Lebesgue measure.

Proof Suppose otherwise. Choose D satisfying

$$\max_i \limsup_{T \rightarrow \infty} \frac{|U_i \cap [0, T]|}{T} < D < C_{r, \lambda}. \quad (4)$$

For all sufficiently large T , therefore,

$$|U_i \cap [0, T]| \leq DT \quad (i \in [r]). \quad (5)$$

No U_i has an unbounded connected component, since that would give density 1. Every component is consequently a finite interval of the form

$$C = [t_f, \lambda t_{\ell+1}],$$

where f and ℓ are the first and last indices of intervals belonging to that component.

For a component C of U_i , define

$$e_C = \lambda t_{\ell+1}, \quad Q_C = \frac{t_{\ell+1}}{t_f}, \quad a_C = \frac{|C|}{e_C} = 1 - \frac{1}{\lambda Q_C},$$

and

$$d_C = \frac{|U_i \cap [0, e_C]|}{e_C}.$$

If C^- is its preceding component of the same color and

$$R_C = \frac{e_C}{e_{C^-}},$$

then

$$d_C = a_C + \frac{d_{C^-}}{R_C}. \quad (6)$$

Discard finitely many components so that all components under consideration have a predecessor and satisfy $d_C \leq D$. For these components,

$$1 - \frac{1}{\lambda} < a_C < d_C \leq D,$$

and thus

$$1 < Q_C < Q_{\max} := \frac{1}{\lambda(1-D)}. \quad (7)$$

Equation (6) gives

$$\log \frac{d_{C^-}}{d_C} = \log R_C + \log \left(1 - \frac{a_C}{d_C} \right) \leq \log R_C + \log \left(1 - \frac{a_C}{D} \right). \quad (8)$$

Consider, on $1 \leq q < Q_{\max}$,

$$F(q) = r \log q + \log \left(1 - \frac{1 - \frac{1}{\lambda q}}{D} \right).$$

For $q > 1$, the inequality $F(q) \geq 0$ is equivalent to

$$D \geq D_{r, \lambda}(q),$$

contrary to $D < C_{r, \lambda}$. Moreover,

$$F(1) < 0, \quad F(q) \longrightarrow -\infty \quad \text{as } q \uparrow Q_{\max}.$$

Continuity therefore supplies $\eta > 0$ such that

$$\log \left(1 - \frac{a_C}{D} \right) \leq -r \log Q_C - \eta \quad (9)$$

for every component under consideration.

Sum (8) and (9) over the retained components whose last index ℓ is at most N . Let their number be m_N .

For each color, the left side telescopes. Since

$$1 - \frac{1}{\lambda} \leq d_C \leq D$$

for retained components, its total is bounded independently of N :

$$\sum_C \log \frac{d_{C-}}{d_C} = O(1). \quad (10)$$

Likewise, telescoping the endpoint ratios gives

$$\sum_C \log R_C \leq r \log t_{N+1} + O(1). \quad (11)$$

The essential covering estimate is

$$\sum_C \log Q_C \geq \log t_{N+1} - O(1). \quad (12)$$

Here is the boundary justification. We have

$$\log Q_C = \sum_{k=f(C)}^{\ell(C)} \log \frac{t_{k+1}}{t_k}.$$

Every index k belongs to the span of its own color-component. The finitely many discarded components account for only a fixed initial contribution. At the right boundary N , at most one component of each color has $f(C) \leq N < \ell(C)$. By (7), the total logarithmic span of each such component is at most $\log Q_{\max}$. Thus the total omitted contribution is bounded independently of N , proving (12).

Combining (8)–(12) yields

$$O(1) \leq r \log t_{N+1} - r \sum_C \log Q_C - \eta m_N + O(1) \leq O(1) - \eta m_N.$$

But $m_N \rightarrow \infty$: there are infinitely many intervals, each finite component contains only finitely many of them, and no component is unbounded. This is a contradiction. $\square$

Evaluation of $C_{r,\lambda}$ Differentiation gives

$$D'_{r,\lambda}(q) = 0 \iff q^r - \lambda r q + r - 1 = 0. \quad (13)$$

The polynomial on the right decreases until $q = \lambda^{1/(r-1)}$, then increases, and is negative at $q = 1$. It therefore has a unique root $b_{r,\lambda} > 1$, satisfying

$$b_{r,\lambda} > \lambda^{1/(r-1)}.$$

This root is the unique minimizer in (2). Since

$$b_{r,\lambda}^r - 1 = r(\lambda b_{r,\lambda} - 1),$$

we obtain

$$C_{r,\lambda} = \frac{b_{r,\lambda}^{r-1}}{\lambda^r}. \quad (14)$$

In particular,

$$\lim_{\lambda \uparrow 2} C_{r,\lambda} = C_{r,2} = \frac{b^{r-1}}{2^r}. \quad (15)$$

2. A restricted-sum fact modulo a prime

The following standard restricted-sum fact is included with a proof so that the additive reduction is explicit.

Lemma 2 Let p be prime and $B \subseteq \mathbb{F}_p$, with $|B| = s$. If

$$1 \leq k \leq s, \quad k(s-k) \geq p-1,$$

then every element of $\mathbb{F}_p$ is a sum of k distinct members of B .

Proof Suppose $c \in \mathbb{F}_p$ is not such a sum. The polynomial

$$P(x_1, \dots, x_k) = ((x_1 + \dots + x_k - c)^{p-1} - 1) \prod_{i < j} (x_j - x_i)$$

vanishes on B^k . Indeed, repeated coordinates annihilate the Vandermonde factor; otherwise the sum differs from c , and Fermat's theorem annihilates the first factor.

Choose distinct integers

$$0 \leq d_1 < \dots < d_k \leq s-1$$

with

$$\sum_{i=1}^k d_i = p-1 + \binom{k}{2}. \quad (16)$$

Such a choice exists because the sums of k distinct integers from $\{0, \dots, s-1\}$ fill the interval

$$\left[\binom{k}{2}, k(s-k) + \binom{k}{2} \right].$$

For example, writing $p-1 = kq + u$, $0 \leq u < k$, one may take $d_i = i-1 + q$ for the first $k-u$ indices and $d_i = i + q$ for the last u .

The coefficient of $\prod_i x_i^{d_i}$ in P is

$$\frac{(p-1)!}{\prod_i d_i!} \prod_{i < j} (d_j - d_i), \quad (17)$$

as follows by expanding the Vandermonde determinant. This is nonzero in $\mathbb{F}_p$, since all $d_i < p$ and the d_i are distinct.

On the other hand, a polynomial of total degree $\sum_i d_i$ with this coefficient nonzero cannot vanish on B^k , because $|B| > d_i$ for every i . This coefficient form of multivariate interpolation follows by choosing sets $X_i \subseteq B$ of size $d_i + 1$: the coefficient equals

$$\sum_{x_i \in X_i} \frac{P(x_1, \dots, x_k)}{\prod_i \prod_{y \in X_i \setminus \{x_i\}} (x_i - y)}.$$

It would be zero if P vanished on B^k , contradicting (17). $\square$

3. Dense prime blocks have long intervals of subset sums

This is the step that removes the restriction to interval-block colorings.

Lemma 3 — Prime-block interval lemma For every fixed $\kappa > 0$, the following holds for all sufficiently large integers n .

If A is a set of primes in $[n, 2n]$ and

$$|A| \geq \kappa \frac{n}{\log n},$$

then

$$\left[n(\log n)^3, \frac{n^2}{(\log n)^3} \right] \cap \mathbb{Z} \subseteq \Sigma(A). \quad (18)$$

Proof Write $m = |A|$. Choose, by Bertrand's postulate, a prime

$$2n < p < 4n.$$

All members of A are distinct modulo p .

Partition A into a group-building set $A^{(0)}$, of size $\lfloor m/2 \rfloor$, and a reserve set $A^{(1)}$. Put

$$h = \left\lceil \frac{8p}{m} \right\rceil = O_\kappa(\log n).$$

For sufficiently large n , we have $h \leq m/8$.

Step 1: Build a long arithmetic progression of subset sums As long as at least $m/4$ elements of $A^{(0)}$ remain, Lemma 2 supplies an h -element subset whose sum is $0 \pmod{p}$. Indeed, if the remaining size is v , then

$$h(v - h) \geq h \frac{m}{8} \geq p.$$

Remove such a subset and continue.

This produces at least $m/(8h)$ disjoint groups. Each group sum has the form

$$\ell p, \quad 1 \leq \ell \leq h,$$

because every group contains h integers between n and $2n < p$. Consequently, for some ℓ , at least

$$L \geq \frac{m}{8h^2}$$

of the groups have the same sum

$$d = \ell p.$$

Their subset sums contain

$$0, d, 2d, \dots, Ld. \quad (19)$$

Here and below 0 is allowed temporarily as an empty sum.

Since $d \geq hn$, the upper endpoint satisfies

$$T := Ld \geq \frac{mn}{8h} \gg_\kappa \frac{n^2}{(\log n)^2}. \quad (20)$$

Step 2: Represent every residue modulo d by a small reserve sum All primes in $A^{(1)}$ exceed ℓ , for sufficiently large n . Thus every occupied residue class modulo ℓ is coprime to ℓ . Choose a residue $c \pmod{\ell}$ for which

$$B := \{a \in A^{(1)} : a \equiv c \pmod{\ell}\}$$

has size

$$s \geq \frac{m}{2\ell} \geq \frac{m}{2h} \gg_\kappa \frac{n}{(\log n)^2}. \quad (21)$$

When $\ell = 1$, the same statements have their usual trivial interpretation.

Set

$$H = \left\lceil \frac{2p}{s} \right\rceil = O_\kappa((\log n)^2).$$

For sufficiently large n ,

$$H + \ell \leq \frac{s}{2}.$$

For every

$$t \in \{H, H + 1, \dots, H + \ell - 1\},$$

we therefore have

$$t(s-t) \geq H \frac{s}{2} \geq p.$$

Lemma 2 says that the sums of t distinct members of B cover all residues modulo p .

Every such sum is also congruent to $tc \pmod{\ell}$. Since $\gcd(c, \ell) = 1$, these ℓ consecutive choices of t cover all residues modulo ℓ . Also $\gcd(p, \ell) = 1$, because $\ell \leq h < p$. The Chinese remainder theorem now shows that **every residue modulo $d = \ell p$** has a representative in $\Sigma(B)$ of size at most

$$M := 2n(H + \ell) = O_{\kappa}(n(\log n)^2). \quad (22)$$

Step 3: Fill an integer interval Let x be any integer with $M \leq x \leq T$. Choose a reserve sum $b_x \leq M$ with

$$b_x \equiv x \pmod{d}.$$

Then

$$x - b_x = qd \quad \text{for some } 0 \leq q \leq L.$$

Use q of the disjoint groups from Step 1, together with the reserve subset representing b_x . These use disjoint elements of A , so

$$[M, T] \cap \mathbb{Z} \subseteq \Sigma(A).$$

Finally, (20) and (22) imply, for sufficiently large n ,

$$M \leq n(\log n)^3, \quad T \geq \frac{n^2}{(\log n)^3}.$$

This proves (18). $\square$

The role of primes is precise: they make the reserve elements units modulo the small integer ℓ . No claim about arbitrary dense integer sets is needed.

4. Applying the geometric lemma to an arbitrary coloring

Let

$$\mathbb{N} = A_1 \sqcup \cdots \sqcup A_r$$

be arbitrary. Put $n_k = 2^k$.

By the prime number theorem, for all sufficiently large k , the interval $[n_k, 2n_k]$ contains at least

$$\frac{n_k}{2 \log n_k}$$

primes. Some color γ_k therefore contains at least

$$\frac{n_k}{2r \log n_k}$$

of these primes. Apply Lemma 3 with $\kappa = 1/(2r)$. We obtain

$$\left[n_k (\log n_k)^3, \frac{n_k^2}{(\log n_k)^3} \right] \cap \mathbb{Z} \subseteq \Sigma(A_{\gamma_k}). \quad (23)$$

Fix $1 < \lambda < 2$, and define

$$t_k = \log n_k + 3 \log \log n_k.$$

For sufficiently large k ,

$$\lambda t_{k+1} \leq 2 \log n_k - 3 \log \log n_k. \quad (24)$$

Indeed, the difference has leading term $(2 - \lambda)k \log 2$, while all remaining terms are $O_\lambda(\log k)$.

It follows from (23)–(24) that every integer whose logarithm lies in

$$J_k = [t_k, \lambda t_{k+1}]$$

belongs to $\Sigma(A_{\gamma_k})$. Set

$$U_i = \bigcup_{\gamma_k=i} J_k,$$

starting at a sufficiently large index.

We now make the comparison with logarithmic density precise. For $E_i = \Sigma(A_i)$, let

$$W_i = \bigcup_{n \in E_i} [\log n, \log(n+1)).$$

For a block J_k assigned to color i , all its points except possibly an initial interval of length at most e^{-t_k} belong to W_i . Hence

$$|U_i \setminus W_i| \leq \sum_k e^{-t_k} < \infty.$$

For every $T > 0$, consequently,

$$\begin{aligned} |U_i \cap [0, T]| &\leq |W_i \cap [0, T]| + O(1) \\ &\leq \sum_{\substack{n \leq e^T \\ n \in E_i}} \log \left(1 + \frac{1}{n} \right) + O(1) \\ &\leq \sum_{\substack{n \leq e^T \\ n \in E_i}} \frac{1}{n} + O(1). \end{aligned} \tag{25}$$

Thus

$$\bar{d}_{\log}(\Sigma(A_i)) \geq \limsup_{T \rightarrow \infty} \frac{|U_i \cap [0, T]|}{T}.$$

Lemma 1 now gives

$$\max_i \bar{d}_{\log}(\Sigma(A_i)) \geq C_{r,\lambda}.$$

This holds for every $\lambda < 2$. Letting $\lambda \uparrow 2$ and using (15),

$$\max_i \bar{d}_{\log}(\Sigma(A_i)) \geq \frac{b^{r-1}}{2r}. \tag{26}$$

The coloring was arbitrary, so

$$c_r \geq \frac{b^{r-1}}{2r}. \tag{27}$$

This is the unrestricted lower bound; no interval-block assumption remains.

5. Matching upper bound

For completeness, the matching construction can also be proved directly.

Let $b > 1$ solve

$$b^r - 2rb + r - 1 = 0,$$

and, for all sufficiently large k , put

$$N_k = \lceil e^{b^k} \rceil.$$

Color

$$[N_k, N_{k+1}) \cap \mathbb{N}$$

by $k \bmod r$, and color the finite initial interval arbitrarily.

If a monochromatic subset sum has its largest summand in block k , then it is at least N_k , and it is less than

$$\sum_{n < N_{k+1}} n < N_{k+1}^2.$$

Therefore, apart from finitely many subset sums,

$$\Sigma(A_i) \subseteq \bigcup_{k \equiv i \pmod{r}} [N_k, N_{k+1}^2]. \quad (28)$$

On the logarithmic scale, these intervals have endpoints

$$b^k + O(e^{-b^k}), \quad 2b^{k+1} + O(e^{-b^{k+1}}).$$

The errors are summable. Harmonic sums over the corresponding integer intervals likewise differ from their logarithmic lengths by a summable error. It is therefore enough to calculate the upper density of

$$V_i = \bigcup_{k \equiv i \pmod{r}} [b^k, 2b^{k+1}].$$

By the minimizer calculation in Section 1,

$$b > 2^{1/(r-1)}.$$

Thus consecutive intervals of the same color are disjoint. At the right endpoint $2b^{k+1}$, their accumulated density tends to

$$\begin{aligned} \frac{\sum_{j \geq 0} (2b-1)b^{k-rj}}{2b^{k+1}} &= \frac{1 - \frac{1}{2b}}{1 - b^{-r}} \\ &= \frac{b^{r-1}}{2r}. \end{aligned} \quad (29)$$

Within an occupied interval the density ratio increases, and within a gap it decreases. Hence (29) is the upper density of V_i .

Equation (28) consequently gives

$$\max_i \bar{d}_{\log}(\Sigma(A_i)) \leq \frac{b^{r-1}}{2r}. \quad (30)$$

Together with (27), this proves the theorem.

6. Constant correction and conclusion

The defining equation implies

$$b^r - 1 = 2rb - r.$$

Therefore

$$\frac{b^{r-1}}{2r} = \left(1 - \frac{1}{2b}\right) \left(1 + \frac{1}{2rb - r}\right).$$

For $r = 2$, $b = 2 + \sqrt{3}$, giving

$$c_2 = \frac{2 + \sqrt{3}}{4},$$

as required.

For $r \geq 3$, however, $2^r > 2r$, so

$$\left(1 - \frac{1}{2b}\right) \left(1 + \frac{1}{2rb - r}\right) < \left(1 - \frac{1}{2b}\right) \left(1 + \frac{1}{2rb - r}\right) = c_r.$$

Thus the catalog formula with 2^r , if interpreted literally, is false even as an upper bound. The theorem proves the intended $2r$ version.

The gap in the previous attempt is closed by Lemma 3: at every dyadic scale, one color contains enough primes to generate an integer interval with logarithmic endpoints

$$\log n + O(\log \log n) \quad \text{and} \quad 2 \log n - O(\log \log n).$$

That is exactly what the geometric argument needs. Arbitrary interlacing of the color classes is therefore allowed, and no unproved reduction to block colorings is used.